MODULE - 2

Stereochemistry

Concept & Types of Isomerism

The organic compounds having same molecular formula but different structure or 3D-plane is called isomer and this phenomena is known as isomerism.

Isomerism

- Structural Isomerism
  - C - Chain
  - P - Position
  - M - Metamerism
  - T - Tautomerism
  - F - Functional
- Stereoisomerism
  - G - Geometrical Isomerism
  - O - Optical Isomerism

1. Structural Isomerism:

The organic compounds which have same molecular formula but different molecular structure is called structural isomer and this phenomena is called structural isomerism.

(i) Chain Isomerism:

M.F. = same
Difference = Chain

C_4H_{10}:

$CH_3-CH_2-CH_2-CH_3$ (n-Butane)

$$\begin{array}{c} CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3 \end{array}$$

(Iso-butane)
(2-Methyl propane)

C_5H_{12}:

$CH_3-CH_2-CH_2-CH_2-CH_3$ (n-pentane)

$$CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-CH_3$$

(Iso-pentane)
(2-methyl butane)

$$CH_3-\overset{\displaystyle CH_3}{\overset{|}{\underset{\displaystyle CH_3}{\underset{|}{C}}}}-CH_3$$

(Neo-pentane)
(2,2-dimethyl propane)

(ii) Position Isomerism:

M.F. = same

Difference = Position of Group

C_3H_7Cl

$CH_3-CH_2-CH_2-Cl$ (1-chloro propane)

$CH_3-\underset{\underset{Cl}{|}}{C}H-CH_3$ (2-chloro propane)

C_3H_8O

$CH_3-CH_2-CH_2-OH$ (1-propanol)

$CH_3-\underset{\underset{OH}{|}}{C}H-CH_3$ (2-propanol)

$C_5H_{10}O$

$CH_3-\overset{\overset{O}{\|}}{C}-CH_2-CH_2-CH_3$ (2-pentanone)

$CH_3-CH_2-\overset{\overset{O}{\|}}{C}-CH_2-CH_3$ (3-pentanone)

(iii) Functional Isomerism:

M.F. = same

Difference = Functional Group

$$C_2H_6O \begin{cases} CH_3-CH_2-OH & \text{(Alcohol)} \\ CH_3-O-CH_3 & \text{(Ether)} \end{cases}$$

$$C_3H_6O \begin{cases} CH_3-CH_2-CHO & \text{(Aldehyde)} \\ CH_3-\overset{\overset{O}{\|}}{C}-CH_3 & \text{(Ketone)} \end{cases}$$

$$C_4H_8O_2 \begin{cases} CH_3-CH_2-\overset{\overset{O}{\|}}{C}-OCH_3 & \text{(Ester)} \\ CH_3-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-OH & \text{(Acid)} \end{cases}$$

(iv) Metamerism Isomerism:

M.F. = same

Difference = No. of carbon b/w hetero-atom.

Heteroatom

$C_4H_{10}O$

$CH_3 - \ddot{O} - CH_2 - CH_2 - CH_3$

$CH_3 - CH_2 - \ddot{O} - CH_2 - CH_3$

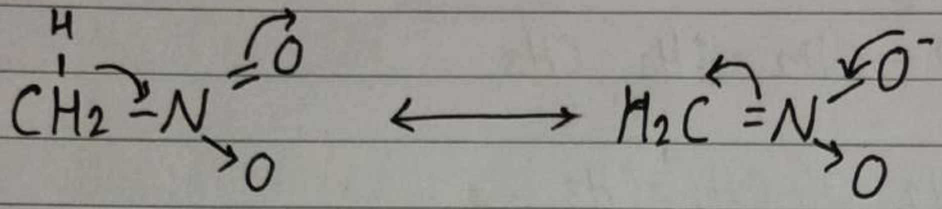
$C_5H_{13}N$

$CH_3 - CH_2 - NH - C_3H_7$

$CH_3 - NH - C_4H_9$

(v) Tautomerism:

The isomers which are in thermodynamic equilibrium are called tautomers and this phenomena is called tautomerism.



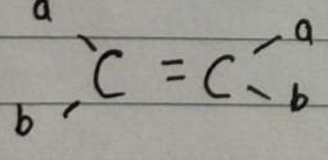
$[O^- \; H^+]$

2. Stereoisomerism:

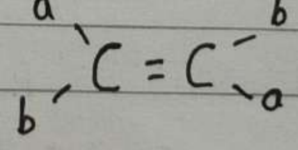
The organic compounds which have same molecular formula but different 3D-plane is called stereoisomer and this phenomena is called stereoisomerism.

(i) Geometrical Isomerism:

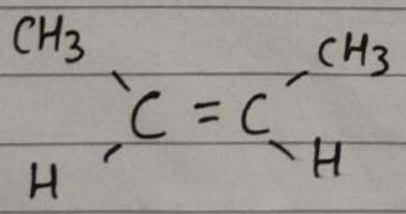
$>C=C<$ [Only]



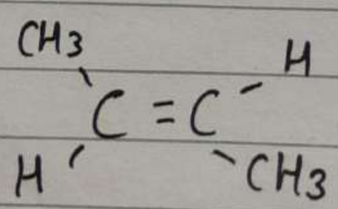
(Cis-isomer)



(Trans-isomer)



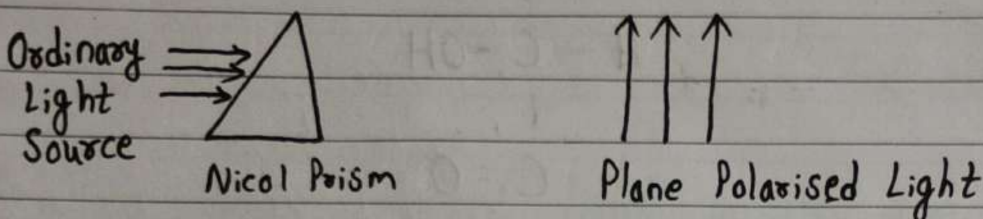
(Cis 2-butane)



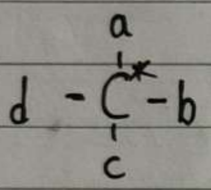
(Trans 2-butane)

Imp
(ii) Optical Isomerism (~~Enantiomerism~~):

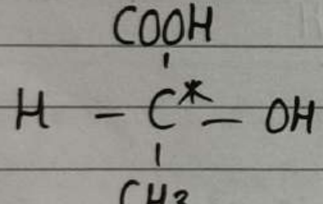
→ Plane Polarised Light



→ Chiral Carbon (Asymmetric Carbon)

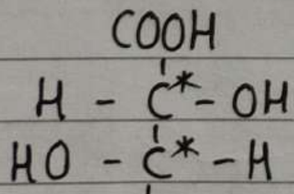


Chiral carbon (n)
No. of optical isomers = 2^n



(Lactic Acid)

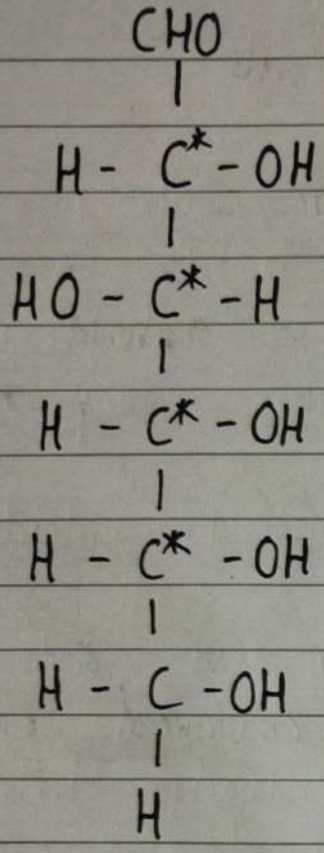
$n = 1$
No. of optical isomers = $2^1 = 2$



$n = 2$
No. of optical isomers = $2^2 = 4$ (Exception)
But structurally only 3 isomers possible.

Glucose (Aldose)

$C_6H_{12}O_6$



$n=4$

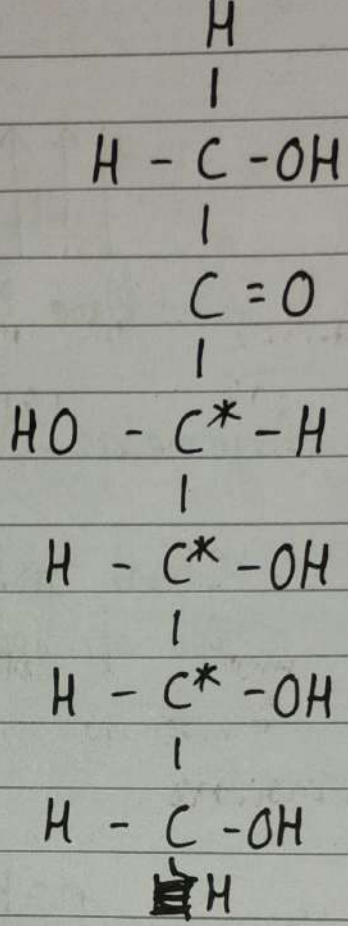
No. of optical isomers

$= 2^4$

$= 16 \left\langle \begin{array}{l} 8d \\ 8l \end{array} \right.$

Fructose (Ketose)

$C_6H_{12}O_6$



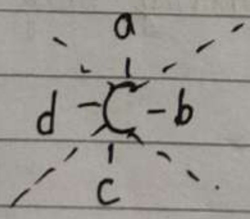
$n=3$

No. of optical isomers

$= 2^3$

$= 8 \left\langle \begin{array}{l} 4d \\ 4l \end{array} \right.$

→ Optical Activity



Asymmetric Carbon

A : A   P : q

Superimposable   Non-superimposable

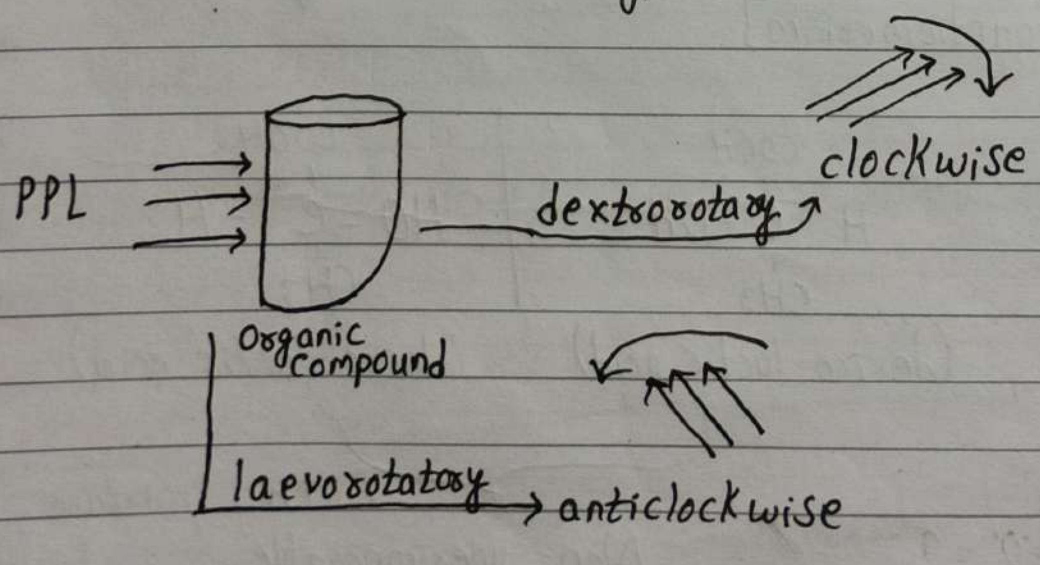
Optically active

→ When plane polarized light passes through the organic compound, if organic compound rotate the PPL into clock-wise direction, then the organic compound is said to be dextrorotatory.

→ Similary, if the organic compound rotate the PPL into anti-clockwise direction, then the organic compound is said to be laevorotatory.

→ If organic compound do not rotate the PPL, then the organic compound is said to be optically inactive.

→ The PPL rotate on angle is called optical active angle.



→ Specific Rotation

$$[\alpha]_D^{25^\circ C} = \frac{\alpha\text{-observed} \times 100}{l \times c}$$

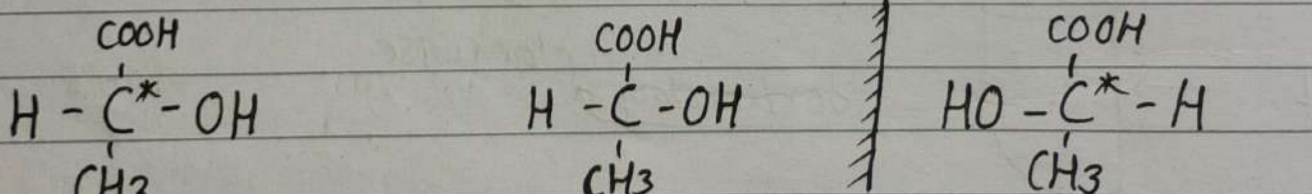
The specific rotation is defined as rotation produced by a solution filled in the 10 cm length tube with concentration of organic compound is 1 gm/ml for the given wavelength of light at the given temperature.

→ Definition of Optical Isomerism:

The stereoisomers which resemble one another in their chemical reaction and different only in their behaviour towards plane polarised light, are called optical isomers and this phenomena is Known as optical isomerism.

Examples

(i) The optical isomerism in compound containing one chiral centre. [Enantiomerism]



(lactic acid) (dextro-lactic acid) (laevo-lactic acid)

Racemic mixture

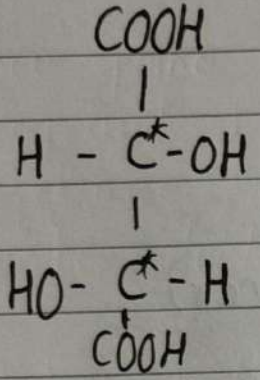
No. of chiral carbon (n) = 1
No. of optical isomers = $2^n = 2$

Non-superimposable
[Enantiomerism]

Enantiomerism: The optical isomers which are non-superimposable mirror image to each other called enantiomers and this phenomenon is known as enantiomerism.

Imp(v)

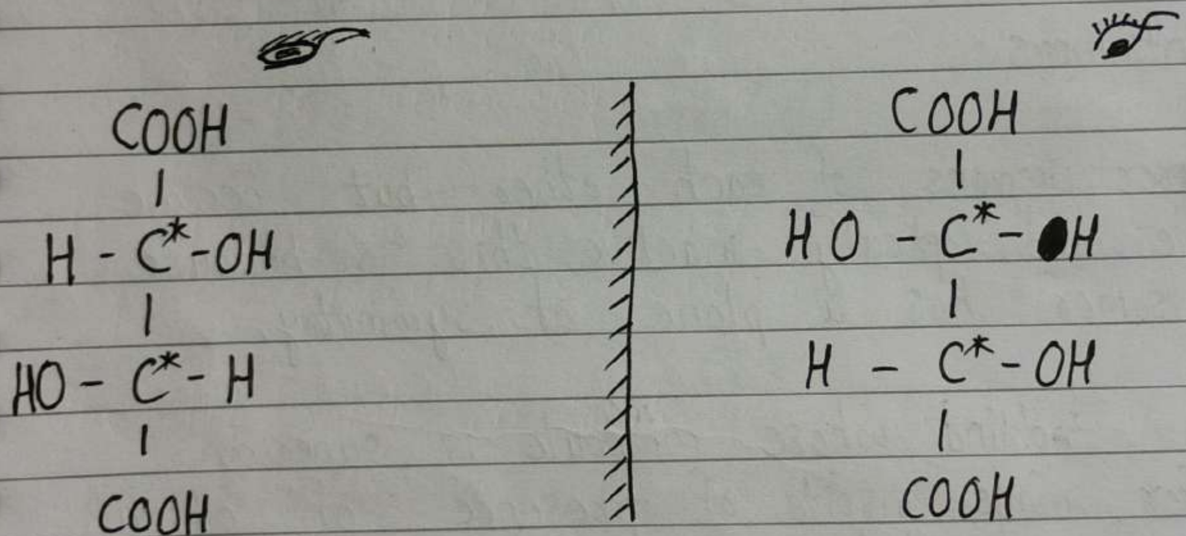
(ii) The optical isomerism in compound containing two similar chiral centre.



Tartaric Acid

Chiral carbon = 2

Optical Isomer = $2^2 = 4$ (3 Optically Active, 1 Optically Inactive)

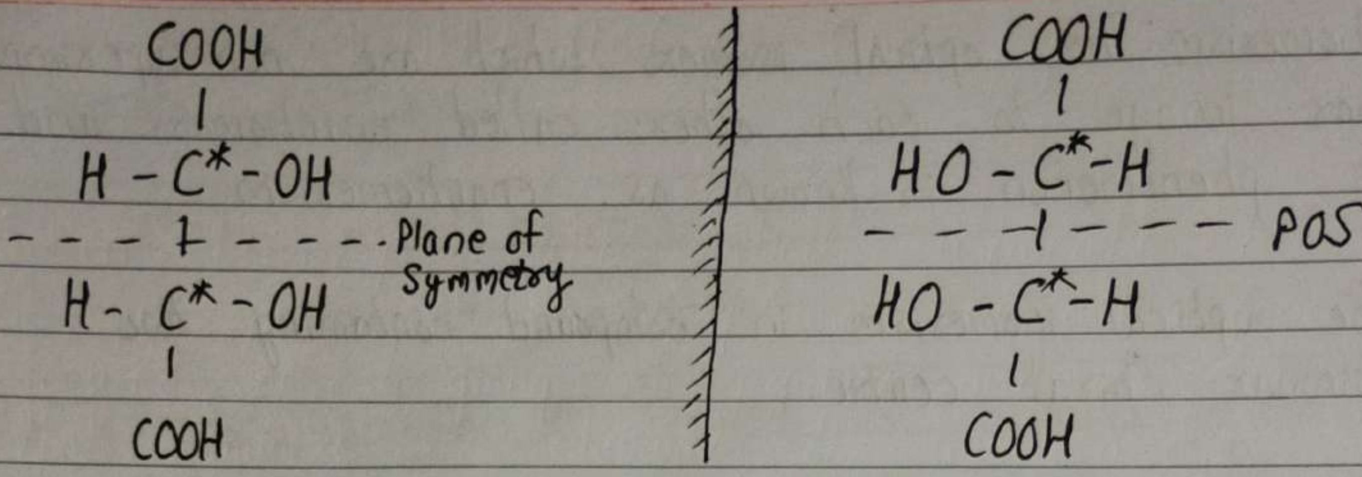


(Dextro-tartaric Acid) [I]

(Laevo-tartaric Acid) [II]

Non-superimposable [Enantiomers]

[Racenic mixture]

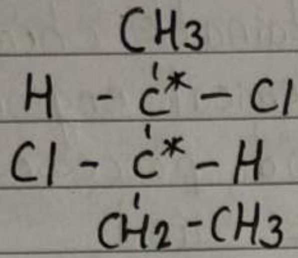


(Meso) [III] [IV]

Superimposable (only one stereoisomer)
Meso optically inactive

- I & II enantiomers.
- III & IV mirror images of each other but become super-imposable and optically inactive, this is because this stereoisomer has a plane of symmetry.
- Such optical compound whose molecule is superim on its mirror image inspite of presence of chiral carbon atom is called meso-compound (optically inactive)
- Optically inactivity of meso compound is due to internal compensation.

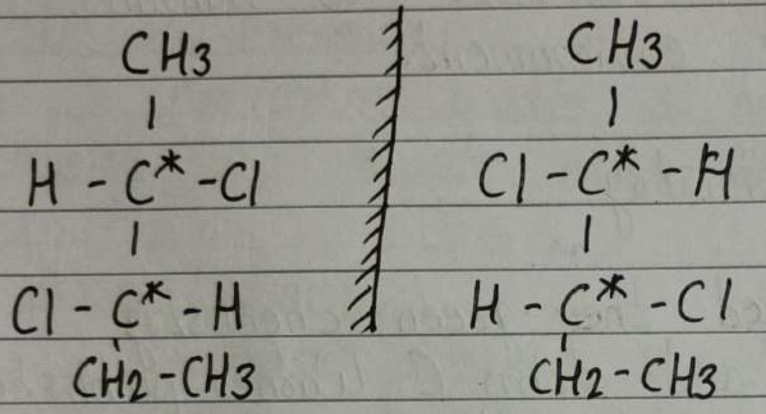
(iii) The optical isomerism in compound containing two disimilar chiral centre.



(2,3-dichloropentane)

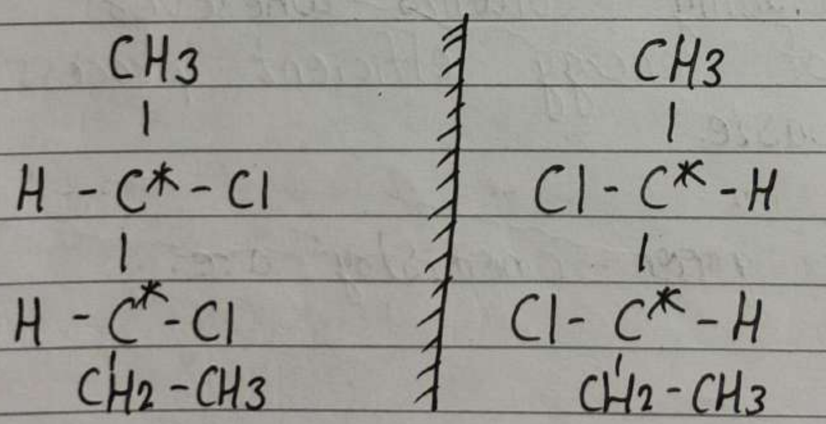
No. of chiral carbon (n) = 2

No. of optical isomerism = $2^2 = 4$



dextro (I) laevo (II)

Non-superimposable [Enantiomers]



dextro (III) laevo (IV)

Non-superimposable [Enantiomers]

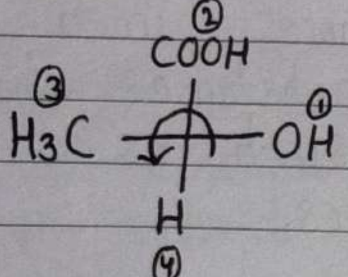
I & II, III & IV → Enantiomers

I & III, I & IV, II & III, II & IV → Diasteromers

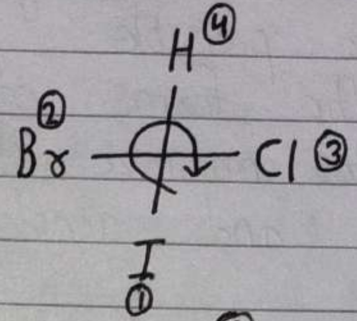
#R-S Nomenclature

R = Rectus (Clockwise)
S = Sinister (Anti-clockwise)

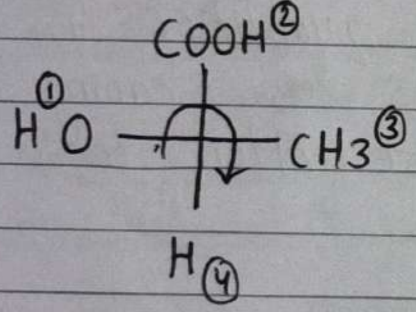
Ist Case: If lowest priority above or below of the plane.



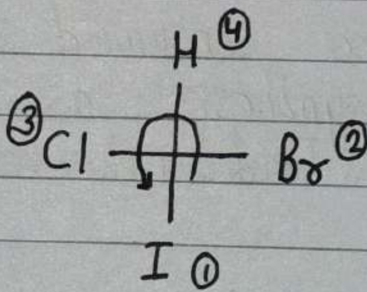
S



R



R



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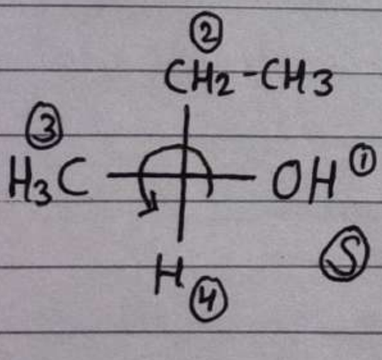
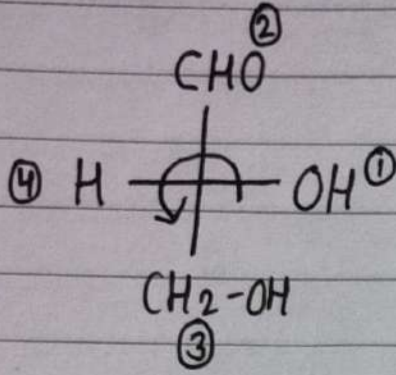


S

II^{nd} Case: If lowest priority left or right of the plane.



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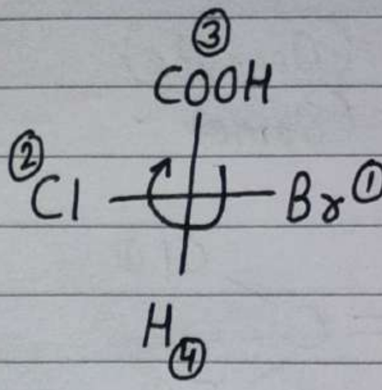

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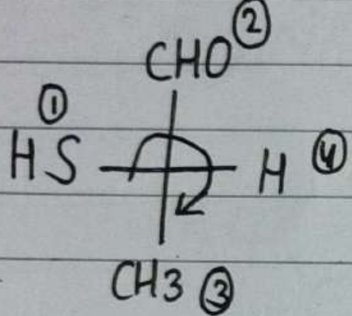

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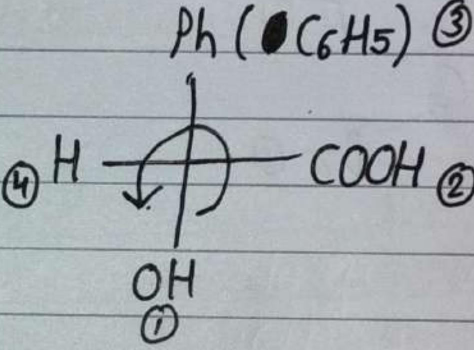

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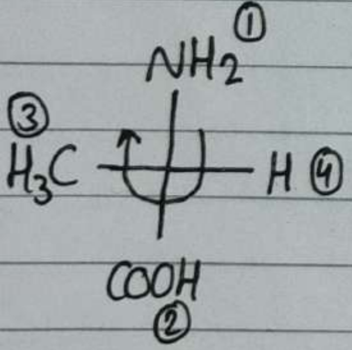

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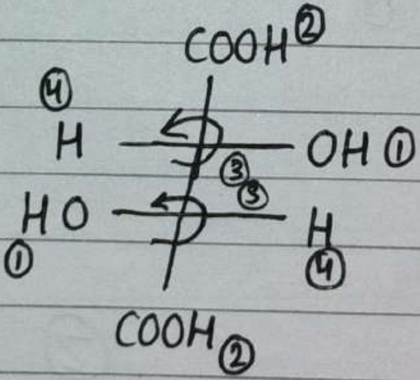

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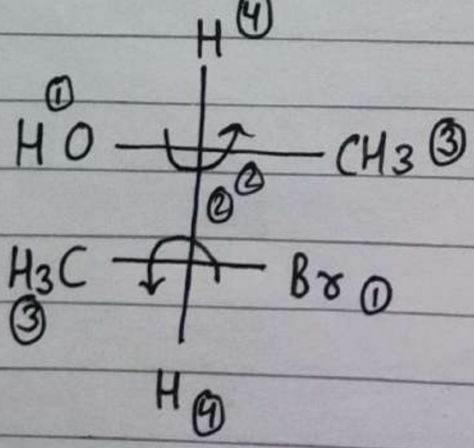

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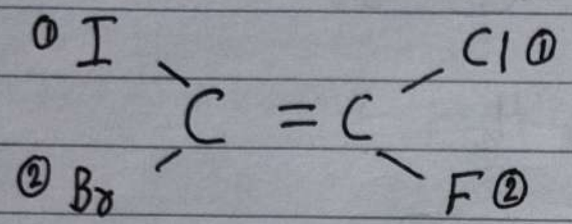
E-Z Nomenclature

E = Entetegen (Opposite)
Z = Znsamann (Same)



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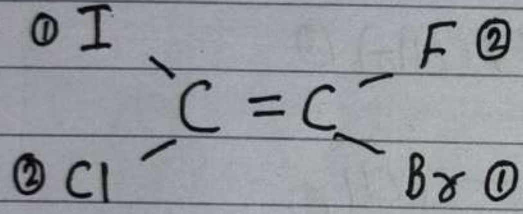

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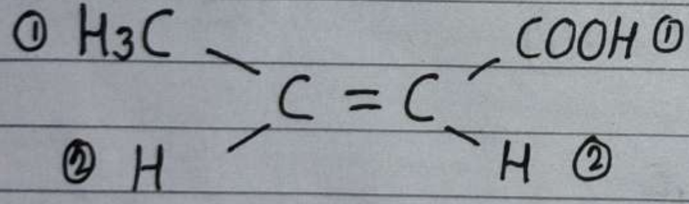

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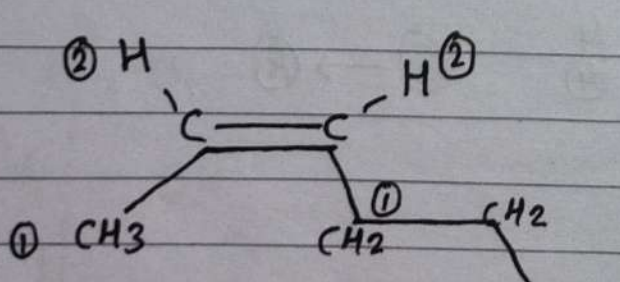

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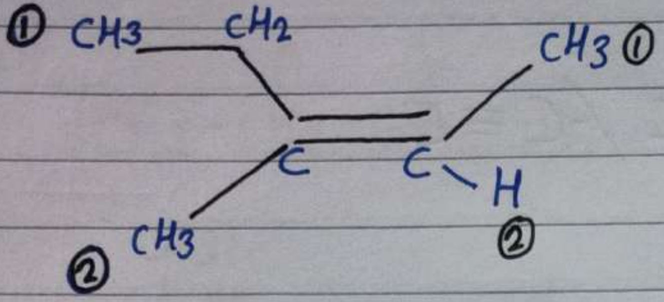
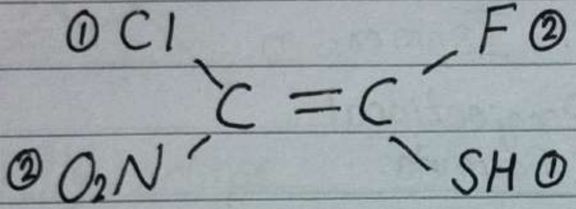
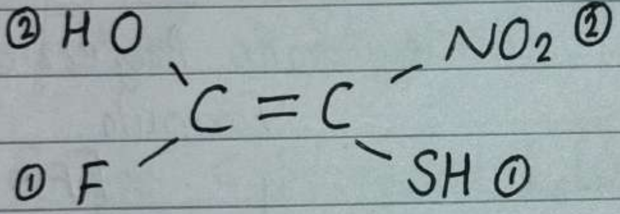


Z



E



Z

Conformational Isomerism

<u>Exception</u>:

$$\begin{array}{c} H \\ | \\ H - C - H \\ | \\ H \end{array} \quad / \; C=C \; / -C \equiv C-$$

(Except CH_4)

Conformational Isomerism

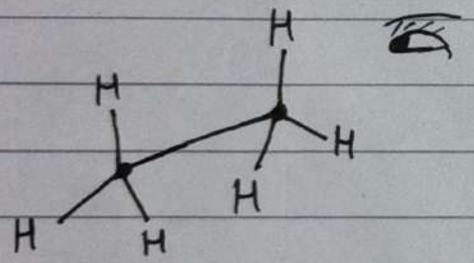
↓ Projectional Formula

Sawhorse Projectional Formula

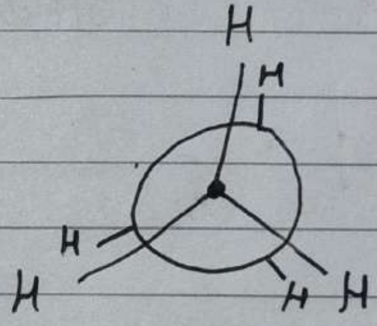
New-mann Projectional Formula

[F → •]

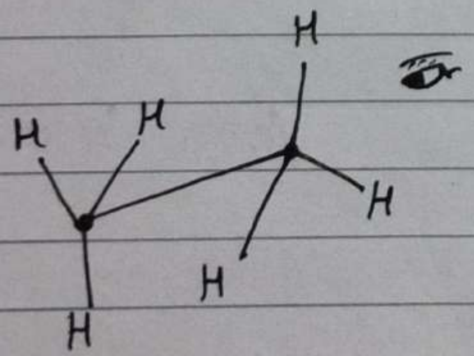
[B → O]



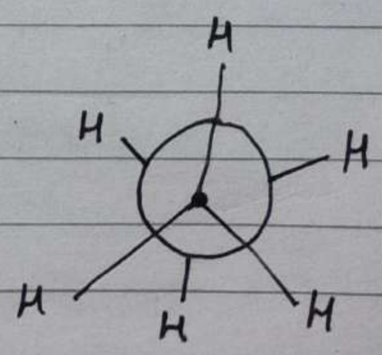
<u>Eclipse</u>



<u>Eclipse</u>



<u>Staggered</u>

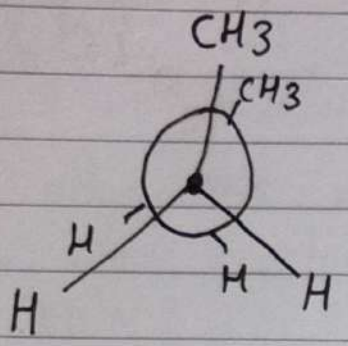


<u>Staggered</u>

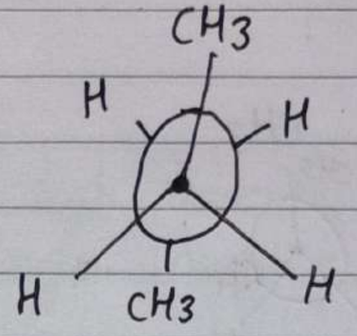
Conformation of n-butane

f b

$CH_3-CH_2-CH_2-CH_3$



Fully Eclipse

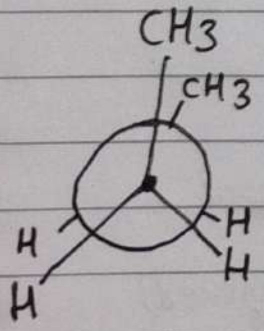


Fully Staggered

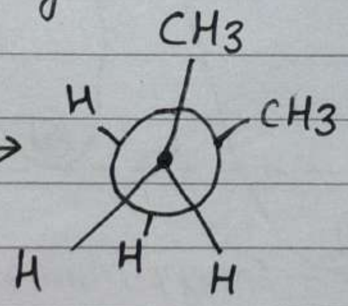
Eclipse > Staggered [Energy]

Eclipse < Staggered [Stability]

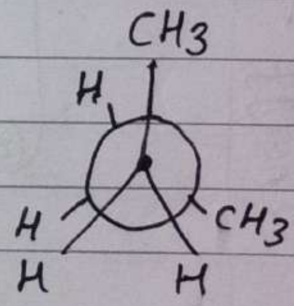
$$\text{Energy} \propto \frac{1}{\text{Stability}}$$



$\xrightarrow{60^\circ}$



$\xrightarrow{60^\circ}$

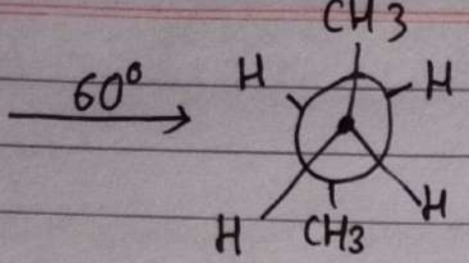


Fully Eclipse [1]

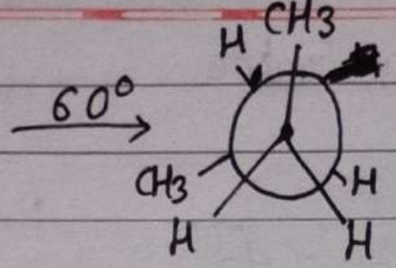
Partial Staggered [2]

Partial Eclipse [3]

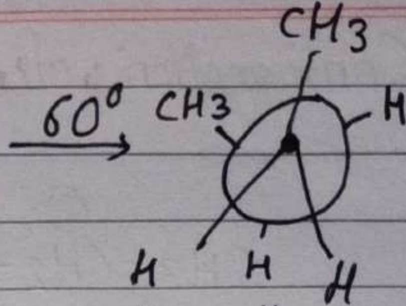
Skew/Gauche Conformation



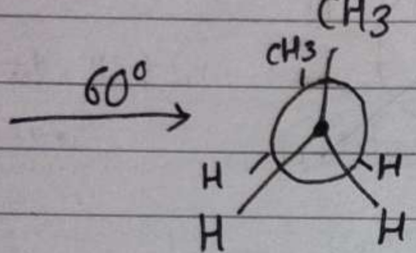
Fully Staggered
[4]



Partially Eclipse
[5]



Partially Staggered
[6]



Fully Eclipse
[7]